

ENTERPRISE SAAS PLATFORM

Enterprise Property Management & Service Marketplace Ecosystem

A comprehensive enterprise-grade digital ecosystem delivering end-to-end property management, maintenance orchestration, and vendor marketplace capabilities — scalable from a single city to nationwide deployment.

Executive Summary

This proposal outlines the architecture and development plan for a **three-tier Enterprise SaaS Ecosystem** designed to digitally transform property management, maintenance operations, and vendor marketplace services — all operating on a single, unified enterprise platform with multi-tenant capabilities.



Understanding of Requirements

ECOSYSTEM STAKEHOLDERS

- Property Owners**
Asset visibility & financial reporting
- Property Managers**
Operational control & tenant management
- Maintenance Teams**
Work order & scheduling management
- Vendors**
Service catalog & order fulfillment
- Technicians**
Field task execution & geo-tracking
- Field Executives**
On-site inspection & reporting
- Central Admins**
Platform-wide governance & analytics
- Platform Admins**
Infrastructure, SLA & tenant control

INDUSTRY CHALLENGES

- ✗ Manual, paper-based operations with no digital audit trail
- ✗ Vendor coordination fragmented across multiple channels
- ✗ Disconnected systems preventing data-driven decisions
- ✗ Delayed maintenance response causing tenant dissatisfaction
- ✗ No centralized monitoring or real-time visibility
- ✗ Poor reporting with no unified analytics layer

PLATFORM SOLUTIONS

- ✓ Fully digitized workflows with end-to-end audit logs
- ✓ Unified vendor marketplace with RFQ & PO management
- ✓ Microservices architecture ensuring seamless data flow
- ✓ Automated maintenance scheduling & SLA tracking
- ✓ Real-time dashboards with Prometheus & Grafana monitoring
- ✓ Advanced analytics with role-based reporting modules

Overall Platform Architecture

CLIENT CHANNELS — USER-FACING APPLICATIONS

- Owner Mobile App
- Vendor Mobile App
- Technician App
- Field Executive App
- PropertyCare Admin
- ServiceHub Admin
- Platform Control Center



API GATEWAY LAYER — APISIX

- Authentication
- Request Routing
- Load Balancing
- Rate Limiting
- SSL Termination
- WAF Protection
- Request Transform



IDENTITY — KEYCLOAK

- SSO
- OAuth 2.0
- OpenID
- RBAC
- MFA
- User Federation

MICROSERVICES — SPRING BOOT (JAVA)

- Property Service
- Maintenance Service
- Vendor Service
- Notification Service
- Analytics Service
- Document Service
- Billing Service



DATA & INFRASTRUCTURE LAYER

- PostgreSQL
Primary Database
- Redis
Cache & Sessions
- RabbitMQ
Message Broker
- MinIO
Object Storage
- Kubernetes
Orchestration
- Prometheus
Monitoring
- Grafana
Dashboards
- Loki
Log Aggregation

PropertyCare SaaS

PropertyCare is a full-featured property lifecycle management SaaS application serving property owners, managers, and tenants through an integrated suite of modules covering the complete operational spectrum.

Property Management

- Asset registry & floor plans
- Occupancy tracking
- Lease management
- Unit configuration

Customer Management

- Tenant onboarding
- Profile management
- Communication history
- Service requests

Maintenance

- Work order creation
- Technician dispatch
- SLA tracking
- Completion reports

Inspection

- Scheduled inspections
- Checklist templates
- Photo documentation
- Defect reporting

Invoices & Billing

- Automated invoicing
- Payment tracking
- Late fee management
- GST compliance

Reports & Analytics

- Occupancy reports
- Revenue dashboards
- Maintenance KPIs
- Trend analysis

Documents

- Agreement storage
- Digital signatures
- NOC generation
- Audit-ready archive

Communication

- In-app messaging
- Email & SMS alerts
- Broadcast notices
- Escalation workflows

ServiceHub SaaS & Platform Control Center

ServiceHub SaaS — Vendor Marketplace

A procurement and vendor management platform connecting property operations with certified service providers, enabling transparent RFQ, PO, and payment workflows.

Vendor Management

- Onboarding & KYC
- Profile & ratings
- Service categories
- Document verification

Catalog & Pricing

- Service catalog
- Price benchmarking
- Item master data
- SKU management

RFQ & Work Orders

- RFQ issuance
- Quote comparison
- Work order dispatch
- Completion sign-off

Payments & Analytics

- Invoice processing
- Payment milestones
- Vendor analytics
- Performance ratings

Platform Control Center — PaaS Admin

A comprehensive platform administration console enabling multi-tenant governance, infrastructure management, license control, and enterprise-grade monitoring.

Tenant Management

- Tenant onboarding
- Subscription plans
- Resource allocation
- Isolation controls

Geo Configuration

- City management
- Zone management
- Region hierarchy
- Coverage mapping

System Operations

- Monitoring & alerts
- Audit log viewer
- Database management
- Backup & restore

Governance

- License management
- System configuration
- User management
- Access policies

Applications Portfolio

Owner Mobile App

Property Owners

KEY FEATURES

- Portfolio dashboard
- Revenue & expense reports
- Maintenance request view
- Document access
- Push notifications

Flutter RBAC
Push

Vendor Mobile App

Service Vendors

KEY FEATURES

- Order management
- Quote submission
- Invoice tracking
- Payment status
- Rating & reviews

Flutter RBAC
Push

Technician App

Field Technicians

KEY FEATURES

- Work order list
- Geo-navigation
- Offline capability
- Photo upload
- Status updates

Flutter RBAC
Push

Field Executive App

On-site Executives

KEY FEATURES

- Inspection checklists
- Defect reporting
- Geo-tracking
- Supervisor reports
- Real-time sync

Flutter RBAC
Push

PropertyCare Admin

Property Managers

KEY FEATURES

- Tenant management
- Maintenance control
- Billing oversight
- Analytics dashboard
- Role-based access

Flutter RBAC
Push

ServiceHub Admin

Procurement Teams

KEY FEATURES

- Vendor directory
- RFQ management
- PO lifecycle
- Payment approval
- Performance KPIs

Flutter RBAC
Push

Platform Control Center

Platform Admins

KEY FEATURES

- Tenant provisioning
- System configuration
- Audit log access
- Backup management
- License governance

Flutter RBAC
Push

Enterprise Technology Stack

LAYER	TECHNOLOGY	PURPOSE & RATIONALE
Backend	Java 21	Primary language — enterprise reliability, performance, ecosystem maturity
	Spring Boot 3	Microservices framework — rapid API development, auto-configuration, production-ready
	Spring Cloud	Service mesh — service discovery, circuit breakers, distributed config
Frontend	React 18	Web admin panels — component-driven UI, virtual DOM, ecosystem breadth
	Next.js 14	SSR & SEO-optimised portals — server-side rendering, static generation
Mobile	Flutter 3	Cross-platform mobile — single codebase for iOS & Android, native performance
Data & Messaging	PostgreSQL 16	Primary RDBMS — ACID compliance, database-per-tenant isolation, JSON support
	Redis 7	Caching & sessions — sub-millisecond latency, pub/sub, rate limiting support
	RabbitMQ	Async messaging — reliable event delivery, dead-letter queues, microservice decoupling
Platform & Security	Keycloak	Identity provider — SSO, OAuth 2.0, OpenID Connect, RBAC, MFA, user federation
	APISIX	API gateway — routing, rate limiting, WAF, SSL termination, plugin ecosystem
Infra & Observability	Docker & Kubernetes	Container orchestration — auto-scaling, rolling deployments, fault isolation
	Prometheus + Grafana	Metrics & dashboards — real-time alerting, SLA monitoring, capacity planning
	Loki + MinIO	Log aggregation + S3-compatible object storage for documents, photos, reports

Security Architecture

Identity & Access Management

- Keycloak-powered SSO across all applications
- Role-Based Access Control (RBAC) with fine-grained permissions
- Multi-Factor Authentication (MFA) for all admin roles
- OAuth 2.0 / OpenID Connect token flows
- User federation with LDAP / Active Directory

API & Transport Security

- End-to-end TLS 1.3 encryption on all channels
- JWT token validation at API gateway layer
- Rate limiting to prevent DDoS & abuse
- Web Application Firewall (WAF) on APISIX
- Input validation & injection prevention

Data Protection

- Database-per-tenant isolation architecture
- AES-256 encryption for sensitive data at rest
- Encrypted MinIO object storage for documents
- Secure signed URLs for file access control
- PII data masking in logs and audit trails

Monitoring & Audit

- Comprehensive audit logs for all user actions
- Real-time anomaly detection via Prometheus alerts
- Centralised log aggregation with Loki
- Security event dashboards in Grafana
- Automated alerting for policy violations

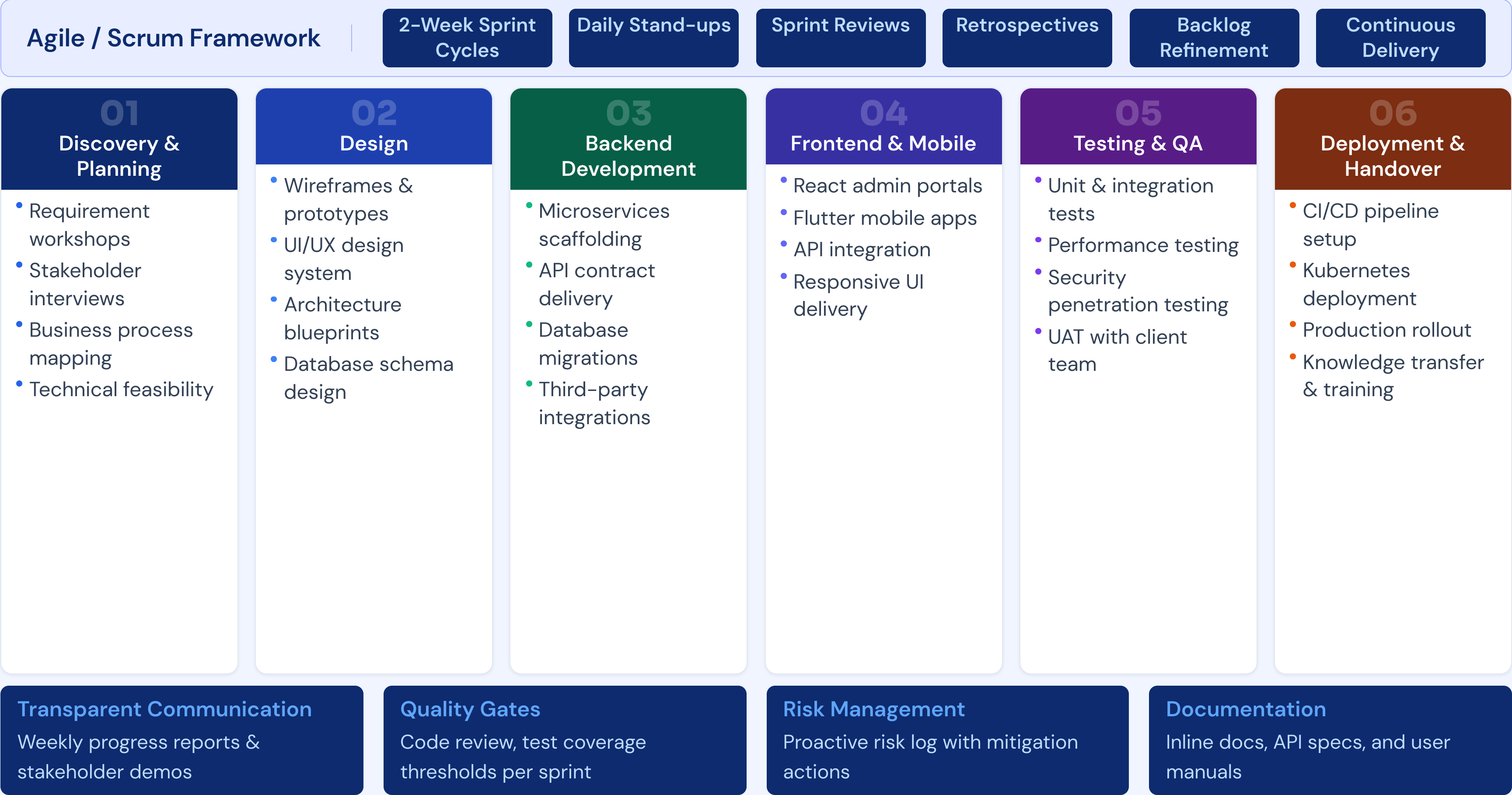
Resilience & Recovery

- Automated daily database backups with encryption
- Point-in-time recovery with configurable RPO/RTO
- Multi-zone Kubernetes deployment for HA
- Disaster recovery runbooks and tested procedures
- Immutable backup storage with retention policies

Compliance & Governance

- Architecture aligned with ISO 27001 principles
- GDPR-ready data processing controls
- Regulatory-grade audit trails for all data changes
- Configurable data residency and geographic controls
- Quarterly security review and penetration testing

Development Methodology



Estimated Project Timeline

Phase	Duration	Scope	Timeline (Weeks 1–20)
Requirement Analysis	2 Weeks	Stakeholder workshops, BRD, gap analysis	
UI/UX Design	3 Weeks	Wireframes, prototypes, design system, approval	
Architecture & Setup	2 Weeks	Infrastructure provisioning, CI/CD, environment setup	
Backend API Development	10 Weeks	Microservices, database, API gateway, integrations	
Mobile Applications (Flutter)	10 Weeks	Owner, Vendor, Technician, Field Executive apps	
Web Portals (React / Next.js)	8 Weeks	PropertyCare Admin, ServiceHub Admin panels	
Platform Control Center	8 Weeks	Tenant management, monitoring, audit console	
Testing & QA	4 Weeks	Unit, integration, performance, security, UAT	
Deployment & DevOps	2 Weeks	Kubernetes deployment, production hardening	
Go Live & Stabilisation	1 Week	Production launch, hyper-care support	
Post-Launch Support	Ongoing	Bug fixes, AMC, SLA-driven maintenance	

8–10

Months Total

Important Disclaimer

All timelines are indicative and subject to detailed requirement analysis, feature freeze, and contractual agreement. Final schedule confirmed post-discovery. Parallel workstreams planned for maximum efficiency.

Rough Cost Estimate

Component	Estimate	Scope Summary
Business Analysis & Consulting	₹3 – 5 Lakhs	Requirements, BRD, process mapping, stakeholder alignment
UI/UX Design	₹5 – 8 Lakhs	Wireframes, prototypes, design system, all 7 application interfaces
Backend API Development	₹20 – 30 Lakhs	All microservices, API gateway, integrations, business logic
Flutter Mobile Applications	₹18 – 25 Lakhs	Owner, Vendor, Technician, Field Executive — iOS & Android
React Web Admin Portals	₹12 – 18 Lakhs	PropertyCare Admin, ServiceHub Admin, Platform Control Center
DevOps & Infrastructure	₹6 – 8 Lakhs	CI/CD pipelines, Kubernetes setup, monitoring & alerting
Testing & QA	₹5 – 8 Lakhs	Unit, integration, performance, security & UAT testing cycles
Deployment & Go-Live	₹3 – 5 Lakhs	Production deployment, hardening, launch support
Documentation	₹2 – 3 Lakhs	Technical docs, user manuals, API documentation
Project Management	₹5 – 8 Lakhs	PM, scrum mastering, reporting, stakeholder coordination
Total Estimated Investment	₹80 Lakhs – ₹1.5 Crore	

TOTAL RANGE

₹80L

to

₹1.5 Cr

Final Cost Depends On

- Detailed scope & feature freeze
- Third-party API integrations
- Infrastructure & hosting plan
- Cloud provider selection
- Support SLA requirements
- Customisation depth

All costs are indicative estimates subject to detailed analysis, final scope, and contractual agreement.

Why Choose Our Team

8+

Years Enterprise Experience

400+

Successful Projects
Delivered

50+

International Clients Served

95%

Client Retention Rate

TECHNICAL EXPERTISE

- | | | |
|-----------------------|------------------------------|-------------------------------|
| • Enterprise SaaS | • ERP & CRM Systems | • Marketplace Platforms |
| • Mobile Applications | • AI & ML Integration | • Cloud Platforms (AWS/Azure) |
| • DevOps & CI/CD | • Microservices Architecture | • API Integrations |
| • Payment Systems | • IoT Integrations | • UI/UX & Design Systems |
| • Kubernetes & Docker | • Reporting & Analytics | • Enterprise Security |

INDUSTRY DELIVERY EXPERIENCE

- Government & Public Sector
- Healthcare & Hospitals
- Education Platforms
- Logistics & Supply Chain
- E-commerce & Retail
- Real Estate & PropTech
- Manufacturing & ERP
- Service Marketplaces
- Business Automation
- Custom ERP Systems

OUR COMMITMENT

Quality

Rigorous code reviews, automated testing, and quality gates at every sprint

Scalability

Architecture designed to scale from single-city to nationwide operations

Security

Enterprise-grade security built into every layer, not bolted on

Performance

SLA-backed response times and load-tested infrastructure

Long-term Support

AMC plans, proactive monitoring, and dedicated support teams

Transparency

Weekly progress reports, live dashboards, open communication

Deliverables & Next Steps

Applications - Flutter Owner Mobile App (iOS & Android) - Flutter Vendor Mobile App - Flutter Technician App - Flutter Field Executive App - PropertyCare React Admin Panel - ServiceHub React Admin Panel - Platform Control Center Portal	Backend & Data - All microservices with documented APIs - PostgreSQL schema & migrations - RabbitMQ event bus configuration - Redis caching layer setup - MinIO object storage integration - Keycloak identity configuration - APISIX gateway configuration	Infrastructure & DevOps - CI/CD pipelines (GitHub Actions) - Docker containerisation for all services - Kubernetes deployment manifests - Prometheus + Grafana monitoring setup - Loki log aggregation configuration - Automated backup & restore procedures - Environment configuration management
Quality & Security - Unit & integration test suites - Performance test reports - Security penetration test report - UAT sign-off documentation - OWASP compliance checklist - Vulnerability remediation report	Documentation & Training - Technical architecture document - API documentation (OpenAPI/Swagger) - Database schema documentation - User manuals for all applications - Administrator guides - Training sessions & knowledge transfer - Video walkthroughs for key workflows	Source Code & IP - Full source code ownership transfer - Git repository with complete history - Signed-off code quality report 90-day post-launch warranty - AMC options (12/24-month plans) - Future enhancement roadmap - AI & IoT integration readiness

Ready to Transform Your Property Operations?

Contact us to schedule a discovery workshop and receive a detailed scope & proposal.

1

Discovery Workshop

2

Detailed Scope

3

Finalised Proposal